

Student Number _____ Name _____ Signature _____

This information will not be visible to the marker.

THE UNIVERSITY
of EDINBURGH**Financial Mathematics**
MATH10003Wednesday 10 May 2023
13:00–15:00**Number of questions: 4**
Total number of marks: 100**PART A** Please complete clearly**Exam Number** _____
as shown on your university card**IMPORTANT PLEASE READ CAREFULLY****Before the examination**

1. Put your university card face up on the desk.
2. **Complete PART A and PART B above.** By completing PART B you are accepting the University Regulations on student conduct in an examination (see back cover).
3. Complete the attendance slip and leave it on the desk.
4. In this examination, candidates are allowed to have three sheets (six sides) of A4 paper with whatever notes they desire written or printed on one or both sides of the paper. Magnifying devices to enlarge the contents of the sheets for viewing are not permitted. No further notes, printed matter or books are allowed.
5. A scientific calculator is permitted in this examination. It must not be a graphic calculator. It must not be able to communicate with any other device.

During the examination

1. Write clearly, in ink, in the space provided after each question. If you need more space then please use the extra pages at the end of the examination script or ask an invigilator for additional paper.
2. **Attempt all questions.**
3. If you have rough work to do, simply include it within your overall answer – put brackets at the start and end of it if you want to highlight that it is rough work.

At the end of the examination

1. This examination script must not be removed from the examination venue.
2. There are extra pages for working at the end of this examination script. If used, you should clearly label your working with the question to which it relates.
3. Additional paper and graph paper, if used, should be attached to the back of this examination script. Write your examination number on the top of each additional sheet.

[Do not write on this page]

1. A company issues a bond with 5 years to maturity and an annual coupon of £100. The face value of the coupon is £1200. The annual continuously compounding risk-free interest rate is 5%.
 - (a) What is the present value of the bond? Please round your answer to 3 decimal places.[4 marks]

If you have used additional space for working then please tick here:

1

- (b) What is the future value of the cash flow generated at six months before maturity? Please round your answer to 3 decimal places. [4 marks]

If you have used additional space for working then please tick here:

□

2. The spot price of a stock S is £60, and it is expected to either increase by 10% or decrease by 5% every period over the next two 3-month periods. The annual continuously compounded risk-free interest rate is 4%.
- (a) i. Verify that this market is free of arbitrage. [2 marks]
- ii. Find the risk-neutral measure $\mathbb{Q}$ by identifying the values of q_u and q_d . [4 marks]
- iii. Draw a diagram to show the evolution of the stock price. [5 marks]

[Space for working continues...]

If you have used additional space for working then please tick here:

- (b) Find the price of a 6-month European **call** option with the strike price £63, written on S . Please round your answer to 3 decimal places. [8 marks]

If you have used additional space for working then please tick here:

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- (c) Find the price of a 6-month American **put** option with the strike price £63, written on S . Please round your answer to 3 decimal places. Is an early exercise optimal? [13 marks]

[Space for working continues...]

If you have used additional space for working then please tick here:

9

- (d) Let us introduce a new type of call option - the fixed-strike lookback call option. Given that the payoff of the lookback call, written on S , with strike price K at maturity T is

$$\max_{n=0,1,\dots,T} \{S_n - K, 0\}.$$

Find the price of a 6-month lookback **call** option with the strike price £63, written on S . Please round your answer to 3 decimal places. [10 marks]

[Space for working continues...]

If you have used additional space for working then please tick here:

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Exam Hall Regulations

The following is a copy of a Notice which is displayed in Edinburgh University Examination Halls for the information of students and staff.

The University of Edinburgh Exam Hall Regulations

1. An examination attendance sheet is laid on the desk for each student to complete upon arrival. These are collected by an invigilator after thirty minutes have elapsed from the start of the examination. Students are not normally allowed to enter the examination hall more than thirty minutes after the start of the examination.
2. Students arriving after the start of the examination are required to complete a "Late arrival form" which requires them to sign a statement that they understand that they are not entitled to any additional time. Students are not allowed to leave the examination hall less than thirty minutes after the commencement of the examination or within the last fifteen minutes of the examination.
3. Books, papers, briefcases and cases must be left at the back or sides of the examination room. It is an offence against University discipline for a student to have in their possession in the examination any material relevant to the work being examined unless this has been authorised by the examiners.
4. Students must take their seats within the block of desks allocated to them and must not communicate with other students either by word or sign, nor let their papers be seen by any other student.
5. Students are prohibited from deliberately doing anything that might distract other students. Students wishing to attract the attention of an invigilator shall do so without causing a disturbance. Any student who causes a disturbance in an examination room may be required to leave the room, and shall be reported to the University Secretary.
6. Personal handbags must be placed on the floor at the student's feet; they should be opened only in full view of an invigilator.
7. An announcement will be made to students that they may start the examination. Students must stop writing immediately when the end of the examination is announced.
8. Answers should be written in the script book provided. Rough work, if any, should be completed within the script book and subsequently crossed out. Script books must be left in the examination hall.
9. During an examination, students will be permitted to use only such dictionaries, other reference books, computers, calculators and other electronic technology as have been issued or specifically authorised by the examiners. Such authorisation must be confirmed by the Registry.
10. The use of mobile telephones is not permitted and mobile telephones must be switched off during an examination.
11. It is an offence against University discipline for any student knowingly
 - to make use of unfair means in any University examination
 - to assist a student to make use of such unfair means
 - to do anything prejudicial to the good conduct of the examination, or
 - to impersonate another student or allow another student to impersonate them
12. Students will be required to display their University Card on the desk throughout all written degree examinations and certain other examinations. If a card is not produced, the student will be required to make alternative arrangements to allow their identity to be verified before the examination is marked.
13. Smoking and eating are not allowed inside the examination hall.
14. If an invigilator suspects a student of cheating, they shall impound any prohibited material and shall inform the Examinations Office as soon as possible.
15. Cheating is an extremely serious offence, and any student found by the Discipline Committee to have cheated or attempted to cheat in an examination may be deemed to have failed that examination or the entire diet of examinations, or be subject to such penalty as the Discipline Committee considers appropriate.